



ATAK

AirAware

Plug-in Version: 0.1

ATAK 5.8.0

08 October 2026



Contents

Overview	4
Before you start	5
The front page	6
The layers	7
A layer's own settings	8
TFR	9
Restrictions in view	10
A restriction's details	11
Airspace	12
Airspace colors and classes	13
Special Use airspace	14
UAS ceilings	15
What a square says	16
Obstacles	17
An obstacle's details	18
METARs	19
Settings	20
The map key	21
UAS Flight Plan	22
The launch point	23
The circle	24
Drawing the area to cover	25
Need a ceiling	26

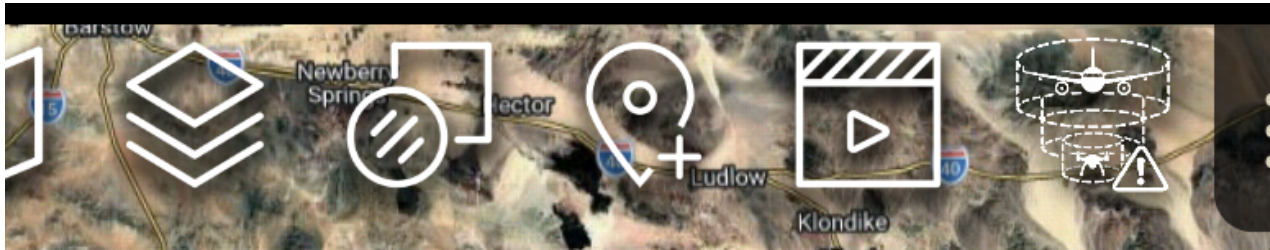
Contents

Given a ceiling	27
Islands	28
Obstacles in the plan	29
The obstacle list	30
An obstacle's details	31
Planner settings	32
Types and the map key	33
When there is no terrain data	34
When there is no network	35
Filing an authorization	36
Notifying Flight Service	37
This manual	38
Worth knowing	39

Overview

AirAware answers one question for anyone working under or in the air: **what is in the column of air I am in, and what am I allowed to do here.**

Six layers draw on the map, each switched on its own. Everything is drawn on this device. Nothing is sent to a TAK Server, nothing is shared with other users, and nothing about you or your position leaves the phone.

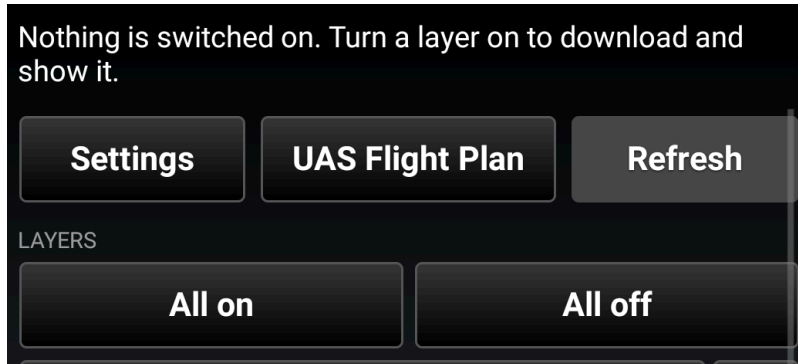


Open it from the ATAK toolbar. The pane opens at half width.

Before you start

- **Everything here is advisory.** AirAware is not a substitute for current charts and NOTAMs, and it is not a clearance. Check the official source before you fly.
- **Network for the first look at an area.** Each layer downloads what it needs for where you are looking, then works from what the phone holds. A phone that has seen an area once can show it again with no signal.
- **Elevation data** is needed by the UAS flight planner only. It reads what ATAK already carries and says so plainly where it is missing.
- **Altitudes.** Restrictions and airspace are drawn between their published floor and ceiling. Obstacle labels are height above the ground; the sea-level top is on the obstacle's own page.

The front page

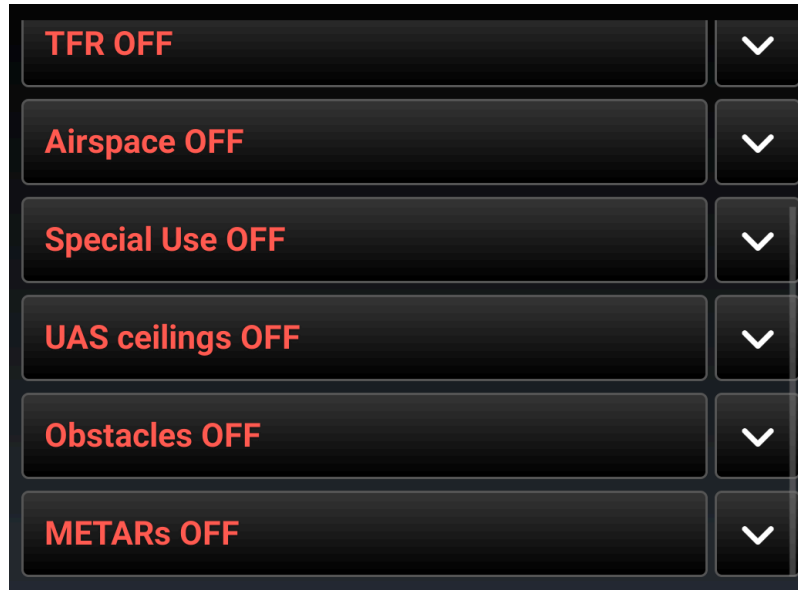


Three buttons across the top:

- **Settings** opens the page that holds everything set once and left.
- **UAS Flight Plan** opens the planner, which is its own screen.
- **Refresh** asks for restrictions and airfield conditions again now.

Under them, **LAYERS**, with **All on** and **All off** for when you want the whole picture or a clean map in one tap.

The layers



One row per layer. The row itself is the switch: it reads the layer's name with **ON** in green or **OFF** in red, and tapping it turns the layer on or off.

Switching a layer off **hides** it. Nothing is deleted, so a layer you turned off before losing signal comes back when you turn it on again.

The arrow on the right opens that layer's own settings, and a line of text saying what it is showing right now. The arrow works while the layer is off, so a layer can be set up before it is switched on.

A layer's own settings

Updated 2 min ago. Names appear at 10 mi or closer.

TFR ON ^

90 restrictions on the map

Types: 6 of 6

Where: Everywhere

Area: What is in view

Zoom gate: Always

Labels: 10 mi

Each layer carries only what belongs to it:

- **TFR:** Types, Where, Area, Zoom gate, Labels.
- **Airspace:** Classes, Show at, 3D, Names on the map.
- **Special Use:** Kinds, Show at, 3D, Names on the map.
- **UAS ceilings:** Show at.
- **Obstacles:** Kinds, Taller than, Show at.
- **METARs:** Show at.

Show at and **Zoom gate** are read off ATAK's own scale bar. Pick the distance the bar has to read before the layer draws, and the layer stays out of the way until you are close enough for it to mean something.

TFR

Temporary Flight Restrictions as the FAA publishes them, drawn as a volume from the published floor to the published ceiling.



Red is in effect now. **Amber** is scheduled and not yet active, so a restriction that starts this afternoon is visible this morning and reads differently from one you are already inside.

Some restrictions are published with no shape at all, as a reference to other airspace. There is nothing to draw for those, and the pane says so rather than leaving them out.

Restrictions in view

RESTRICTIONS IN VIEW (2)

DISNEYLAND THEME PARK, A...
surface to 3,000 ft AGL 3,000 ft above
the ground
Security ACTIVE
tap to go there

Details

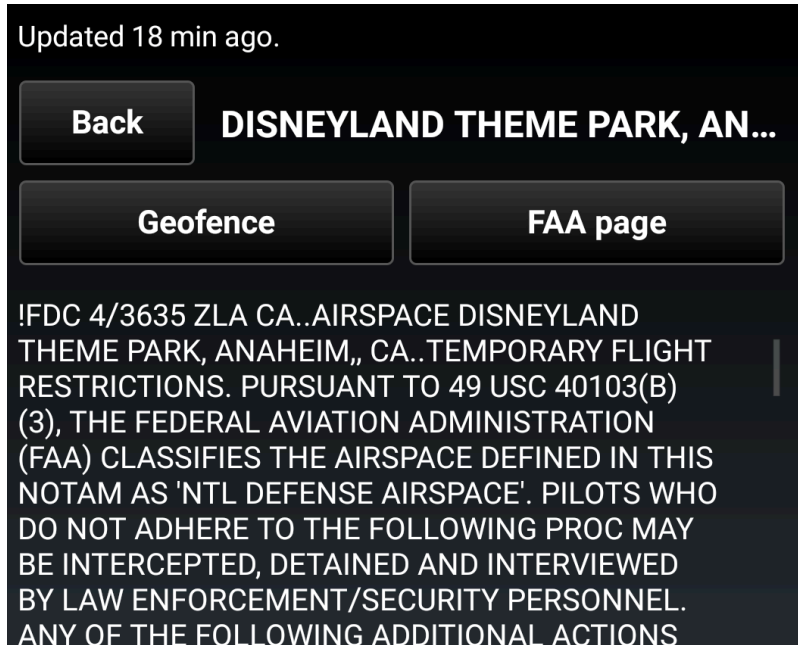
16.71 mi

Apple Valley
surface to 10,000 ft MSL 8,882 ft
above you
Air shows and sports SCHEDULED
tap to go there

Under the layers is the list of what is on the map, newest first. Each row says what the restriction is and when it runs.

The list and the map follow one rule. If a restriction is not on the map it is not in the list, and the status line at the top says why: the zoom gate, the type filter, or that nothing has reached the phone for this area yet.

A restriction's details



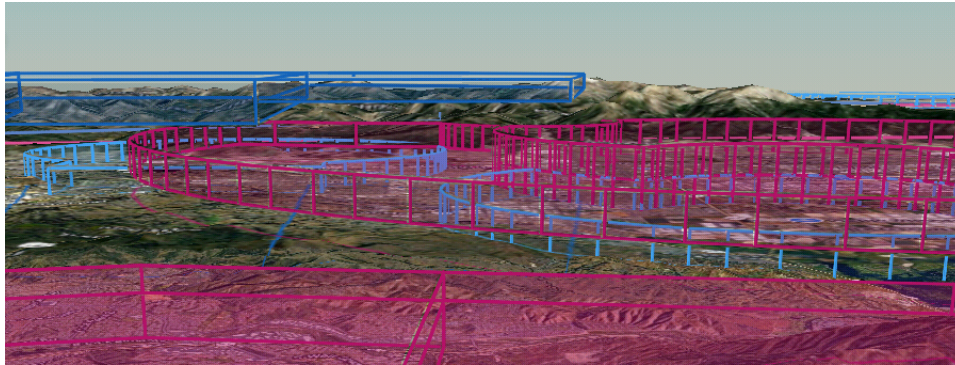
Tap a row, or tap the shape on the map, for the restriction's own page: what it is, where, the floor and ceiling, when it runs, and the full text as the FAA wrote it.

FAA page opens the official notice in a browser.

Geofence puts an ATAK geofence on the restriction, so you are warned when you cross it. Geofences you have made are listed under Settings, and the pane tells you when one no longer matches a live restriction.

Airspace

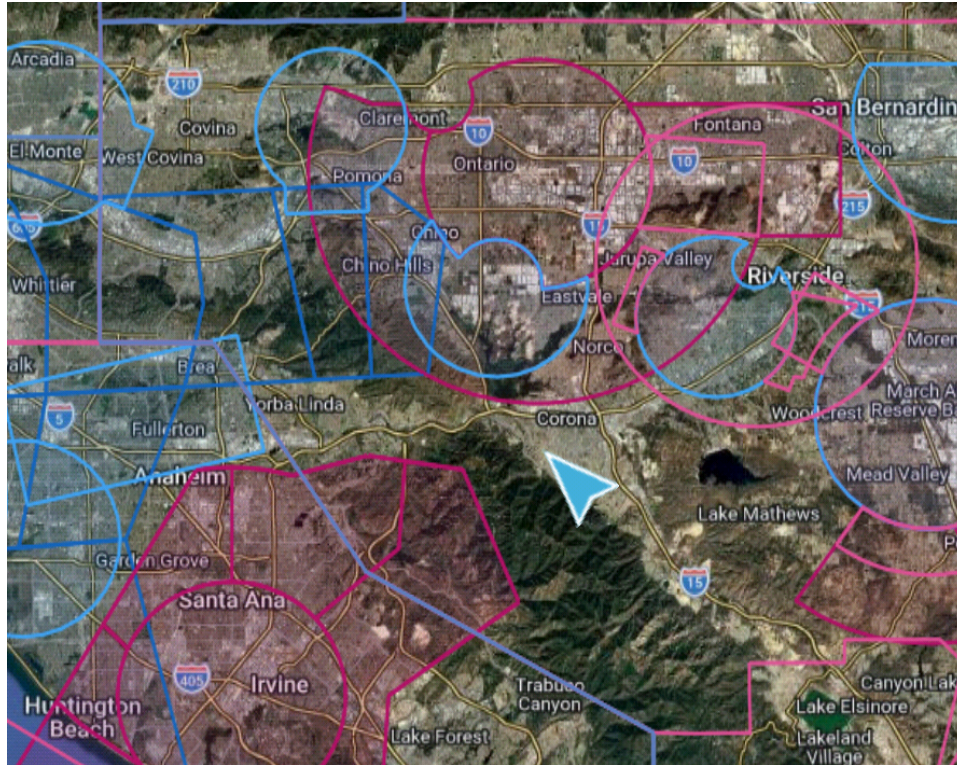
Class A through G, drawn one shelf at a time.



The FAA publishes a Class B airport as a stack of rings, each starting and ending at a different altitude. AirAware draws each of those steps as its own volume, so the upside-down wedding cake is the shape it really is rather than one ring with a note.

With **3D** on, each shelf stands between its own floor and ceiling. Tilt the map to see it. Airspace that goes up with no published ceiling is drawn so it does not swallow everything below it.

Airspace colors and classes

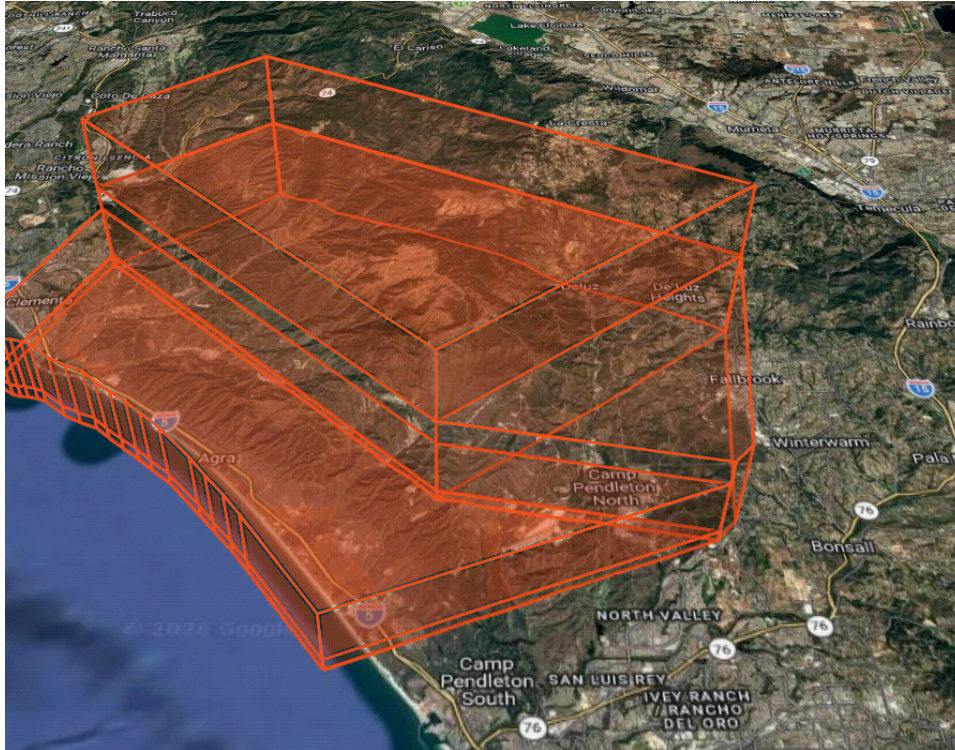


The colors are the sectional chart's: **blue** for Class B and D, **magenta** for Class C and E. Class E takes its color from how low it comes down, so Class E starting at 1,200 ft above the ground reads differently from Class E at 700 ft.

Classes under the layer picks which ones draw. On a busy map, turning off the class you are not working under is the fastest way to see what matters.

Names on the map puts each shelf's name and heights beside it.

Special Use airspace

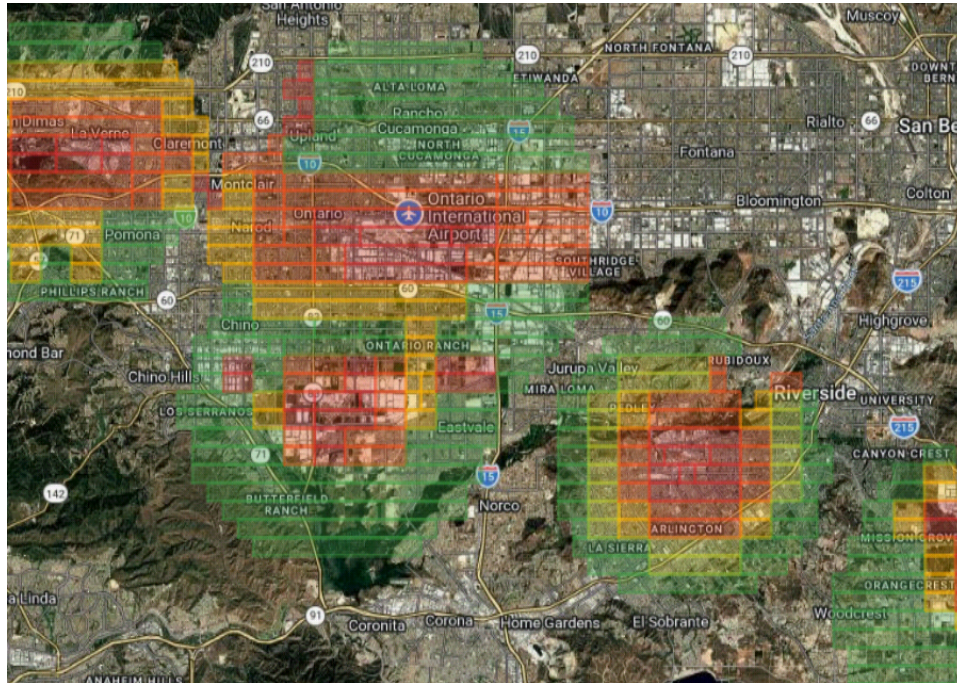


Prohibited, Restricted, Warning and Alert areas, and Military Operations Areas. **Kinds** picks which of those draw.

Each area's page says whether it is continuous or activated by NOTAM. An area that is only active when a NOTAM says so is still drawn, because you need to know it is there before you go looking for the NOTAM.

UAS ceilings

How high a drone may fly without an authorization, from the FAA's UAS Facility Map.



A grid of squares colored by their ceiling: **green** where you have the full height, through to **red** where you have very little. A **0 ft** square is drawn hardest of all, because it is the one that means you cannot fly there without coordinating first.

The grid is flat on purpose. It is a rule about the ground under you, not a volume in the air.

What a square says

Back **200 ft AGL**

Airspace of: RAL Riverside Muni

LAANC: covered - file in an app and it comes back in seconds

The square is 30 arc-seconds, about half a nautical mile. The grid covers controlled airspace only: where there is no square you are in Class G and need no authorization.

FAA UAS Facility Map. It says what is permitted, not what is there - check the restrictions, airspace and obstacle layers too, and always current charts and NOTAMs.

Tap a square for its own page: the ceiling in feet above the ground, the airport the square belongs to, and whether that airport is covered by LAANC.

That last line is the difference between filing in an app and filing for further coordination, which is why it is on the page rather than left to be looked up.

These are the heights an authorization is granted against. They are not a clearance by themselves.

Obstacles

The FAA's Digital Obstacle File: towers, wire spans, transmission lines, turbines and stacks.



Each obstacle stands from the ground to its published top, with the tower symbol at the top and its height on a label above that.

The label is height **above the ground**, because that is the number somebody flying low compares against their own height above the ground.

Taller than and **Kinds** keep the map readable. Over a city the file holds thousands, so the plugin draws the ones nearest what you are looking at and says in words when it has trimmed the rest.

An obstacle's details

[Back](#)

Transmission tower

79 ft above the ground
786 ft MSL at the top

RIVERSIDE, CA

Lighting: N
Survey: unverified
FAA number: 06-242489

You are 468 ft above its top.

FAA Digital Obstacle File. Charting obstacles only - not every tower, and no distribution lines.

Tap an obstacle for its page: height above the ground and its top above sea level, its lighting, whether the FAA has verified the position, how many there are at that point, and the nearest town.

Not every wire or tower is charted. The FAA's file holds the obstacles that affect charting; distribution lines are never in it. Treat an empty map as an absence of records, not an absence of wire.

METARs

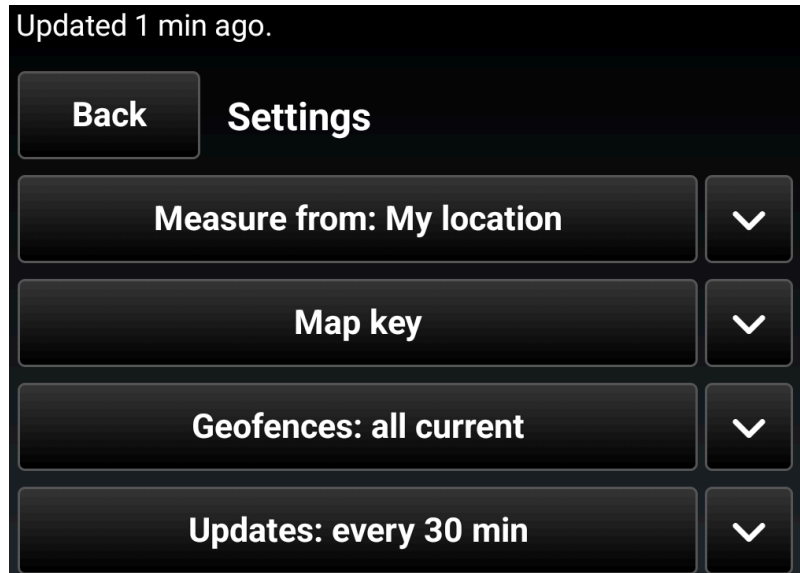


METAR observations drawn as flight-category chips in the aviation convention: **green** VFR, **blue** MVFR, **red** IFR, **magenta** LIFR.

The category is the one published with the observation. It is never worked out here from the numbers, so what you see is what the station reported.

Tap a chip for the observation itself and the time it was made. These are the one thing AirAware does not keep: an old observation is worse than none, so conditions are shown only while they are current.

Settings

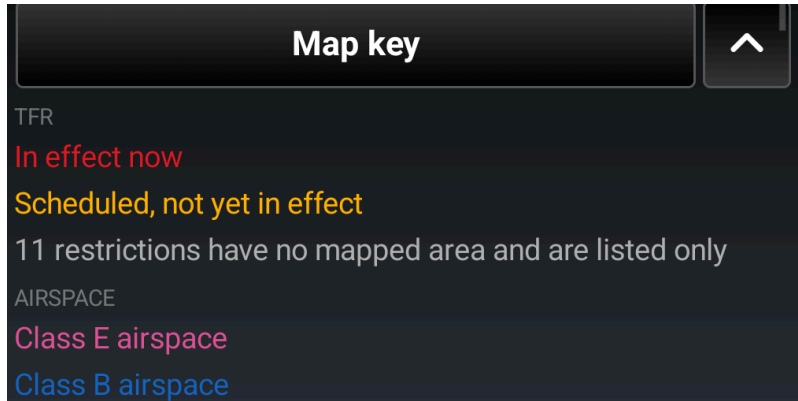


Everything set once and left, on folds that stay as you leave them:

- **Measure from** decides what distances are measured from.
- **Map key** names every color on the map.
- **Geofences** lists the ones you have made, and offers to remove a geofence whose restriction is no longer live.
- **Updates** is how often restrictions and airfield conditions are asked for again.

At the bottom, a line saying what this phone has already downloaded.

The map key

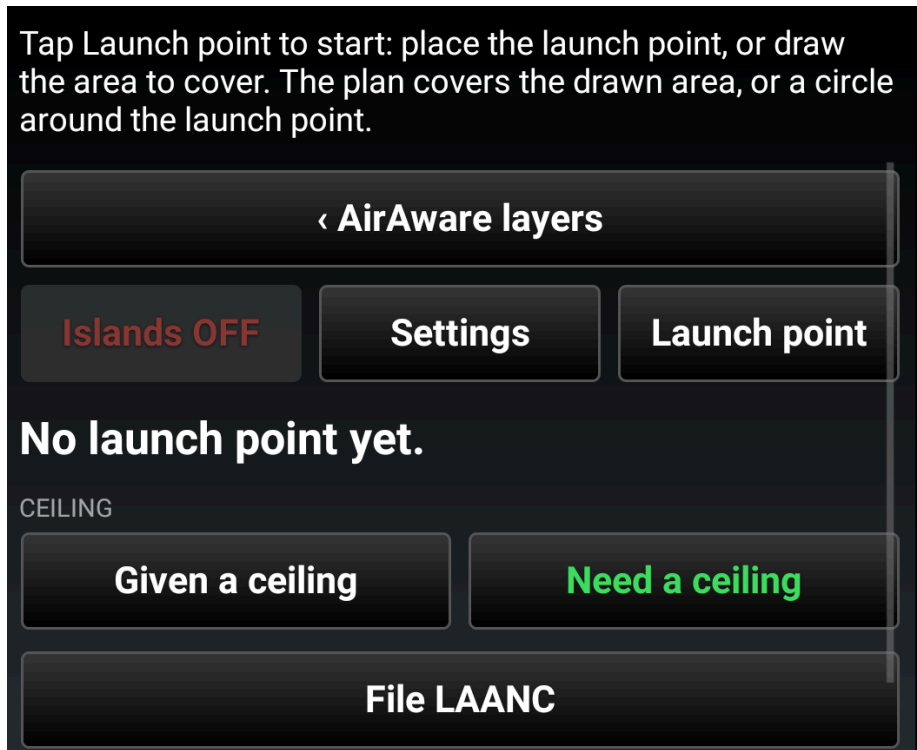


Open **Map key** under Settings for what every color means, in the same words the rest of the pane uses.

The key follows the same rule as the list and the map: if a layer is off, its colors are not in the key.

UAS Flight Plan

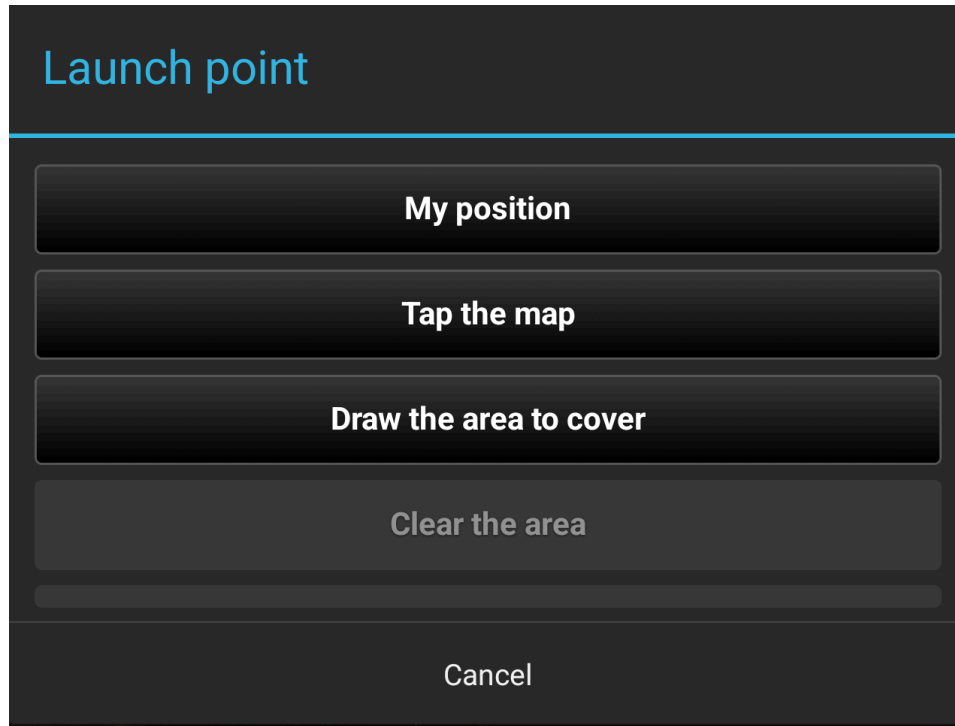
The planner is its own screen, reached from **UAS Flight Plan** on the front page. It answers two questions a UAS pilot has at the turnout before the radio call to Air Attack: **what ceiling does this mission area need**, and **what does the ceiling I was given do to it**.



At the top, **AirAware layers** goes back to the layer switches. Under it the status line, then **Islands ON/OFF**, **Settings** and **Launch point**, and at the bottom **File LAANC** and **Notify Flight Service**.

The planner keeps its own map drawings, separate from the layers.

The launch point

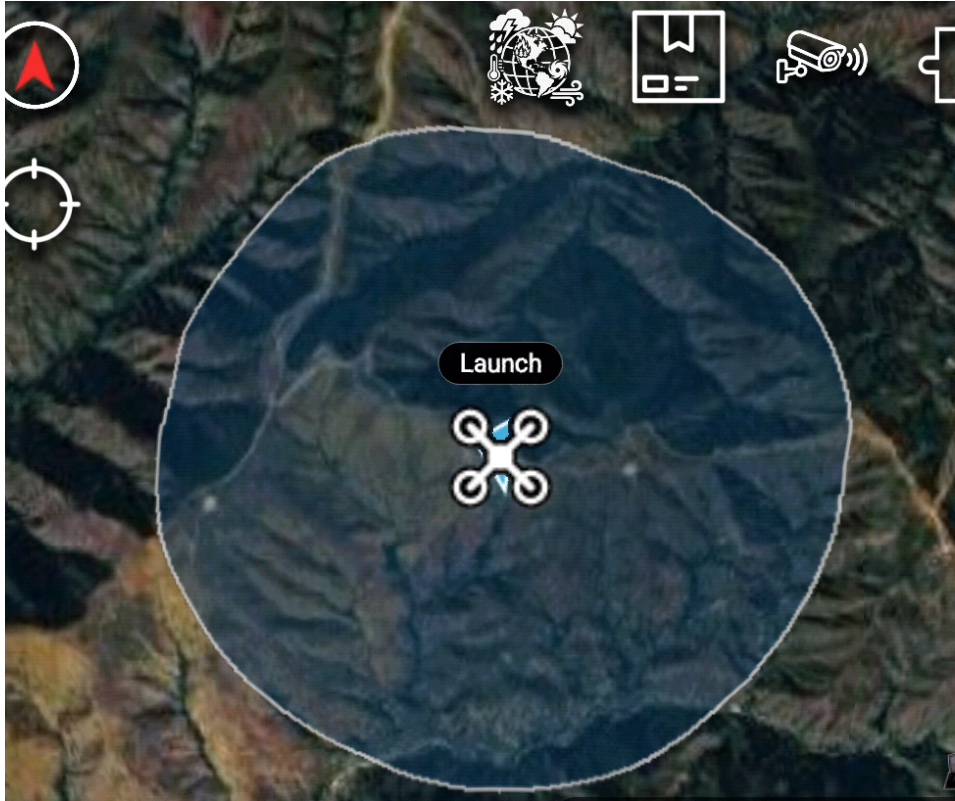


Tap **Launch point**. **My position** uses your own GPS fix; **Tap the map** lets you place it anywhere. **Draw the area to cover** starts the area instead.

The plan covers the drawn area if there is one, otherwise a circle around the launch point.

Everything you set here comes back when ATAK restarts, with no network.

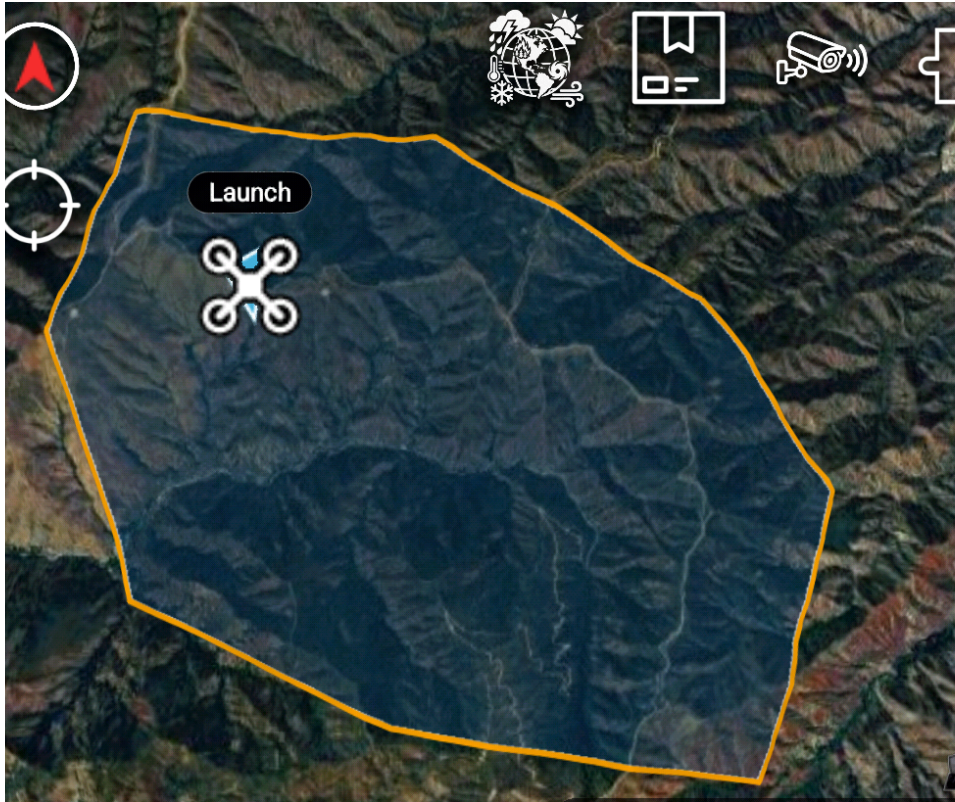
The circle



With no drawn area, the plan is a circle around the launch point. Set its size under Settings.

The circle is what the terrain and the obstacles are read for, so a bigger circle is a higher ceiling and a longer read.

Drawing the area to cover



Draw the area to cover hands you ATAK's own drawing tools. Tap the map to place each corner, then tap the first marker to close the shape. Undo and End Shape are on ATAK's drawing toolbar.

The plan then covers exactly that shape rather than a circle, which matters when the mission is a strip along a ridge.

Need a ceiling

Under **Ground here** is the **Ceiling** row with two buttons. Pick the one that is your situation; the chosen one is green.

All the ground in the area can be worked over. Highest ground in the area: 5,687 ft MSL.

Circle 3 mi, terrain checked every 62 ft.

2 obstacles in the area, 0 above the ceiling.

CEILING TO ASK FOR

Highest ground in the area: 5,687 ft MSL

Fly above the terrain: + 400 ft

Ask Air Attack for 6,100 ft MSL (rounded up to the next 100 ft)

That is 2,523 ft above your launch point: the number for the controller.

OBSTACLES IN THE AREA, NEAREST TO YOU FIRST (2)

Tower, 92 ft AGL

top at 5,701 ft MSL, unlit,

19.17 mi WNW

Details

You are planning and have to ask Air Attack for a ceiling. The block reads the highest ground in the area, adds the height you fly above the terrain (Settings, Height above terrain), and rounds up to the next 100 ft.

The last line is the same number above your launch point, which is what the controller wants.

Given a ceiling

Islands cover 14% of the area. Highest ground in the area: 5,687 ft MSL.

Circle 3 mi, terrain checked every 62 ft.

2 obstacles in the area, 2 above the ceiling.

WHAT THAT CEILING DOES TO THIS AREA

Fly above the terrain: 400 ft

Highest ground you can work over: 4,100 ft MSL

Highest ground in the area: 5,687 ft MSL, 1,587 ft too high. The red islands cannot be worked at this height above the terrain.

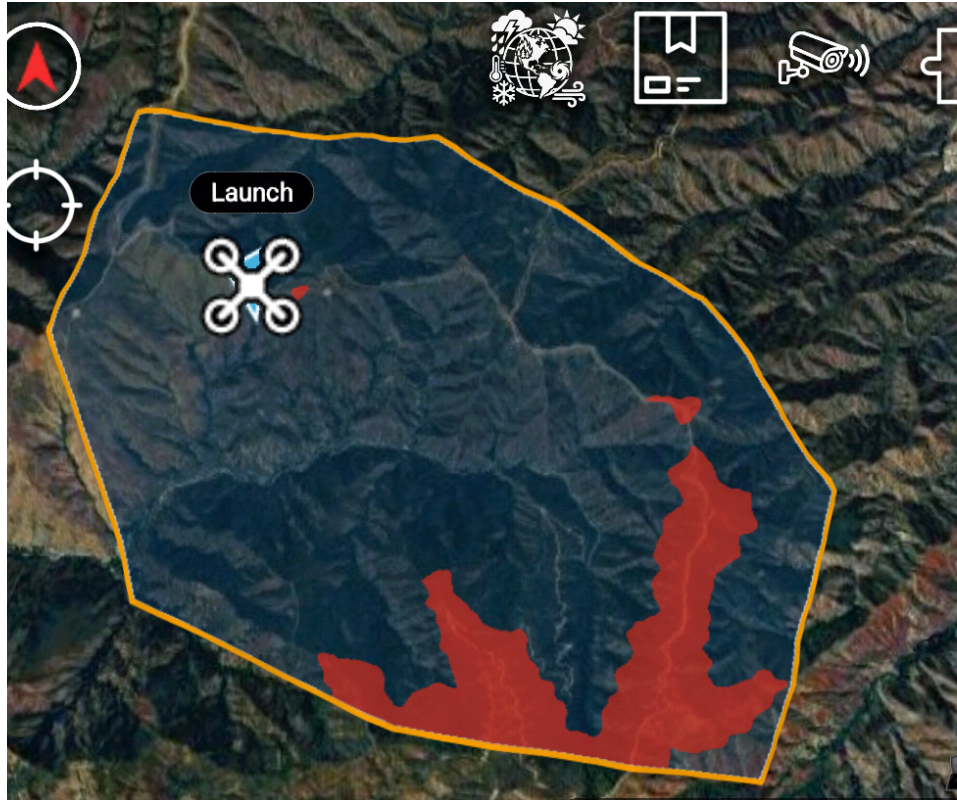
That is 923 ft above your launch point: the number for the controller.

OBSTACLES IN THE AREA, NEAREST TO YOU FIRST (2)

Air Attack assigned you a ceiling and you have to live with it. Tap **Given a ceiling** and type it. The block reads what that ceiling does to your area: the highest ground you can work over at your height above terrain, and how much of the area is too high.

The mission area has to be blue, or the ceiling is wrong for the mission.

Islands



Red islands are the ground you cannot work over at your height above the terrain. Blue is ground under the ceiling.

The edge of the painted area is the edge of what was checked, not a boundary in the world.

Obstacles in the plan



The planner draws the charted obstacles inside the plan area and says how many of them are above your ceiling.

These come from the same FAA file as the Obstacles layer, read from what the phone already holds, so the planner works with no signal.

The obstacle list

OBSTACLES IN THE AREA, NEAREST TO YOU FIRST (2)		
Tower, 92 ft AGL top at 5,701 ft MSL, unlit, verified, CORONA	19.17 mi WNW	Details
Tower, 124 ft AGL top at 5,807 ft MSL, unlit, verified, MISSION VIEJO	19.26 mi WNW	Details

Under the readouts, the obstacles in the area, nearest first, each with its height, its top above sea level, its lighting and how far away it is.

An obstacle's details

All the ground in the area can be worked over. Highest ground in the area: 5,687 ft MSL.
Circle 3 mi, terrain checked every 62 ft.
2 obstacles in the area, 0 above the ceiling.

Back

Tower, 92 ft AGL

Zoom to

FAA Digital Obstacle File

Height above ground: 92 ft

Top: 5,701 ft MSL

Ceiling 6,100 ft MSL: the top is 399 ft under it.

Lighting: unlit

Verified by the FAA: yes

Near: CORONA, CA

From view 10-16 mi

Details opens one obstacle: its heights, whether its top is above or under your ceiling, and the distance and bearing from the launch point.

Planner settings

All the ground in the area can be worked over. Highest ground in the area: 5,687 ft MSL.
Circle 3 mi, terrain checked every 62 ft.
2 obstacles in the area, 0 above the ceiling.

Back

Settings

Obstacles ON



Taller than: 50 ft



Types: 6 of 9

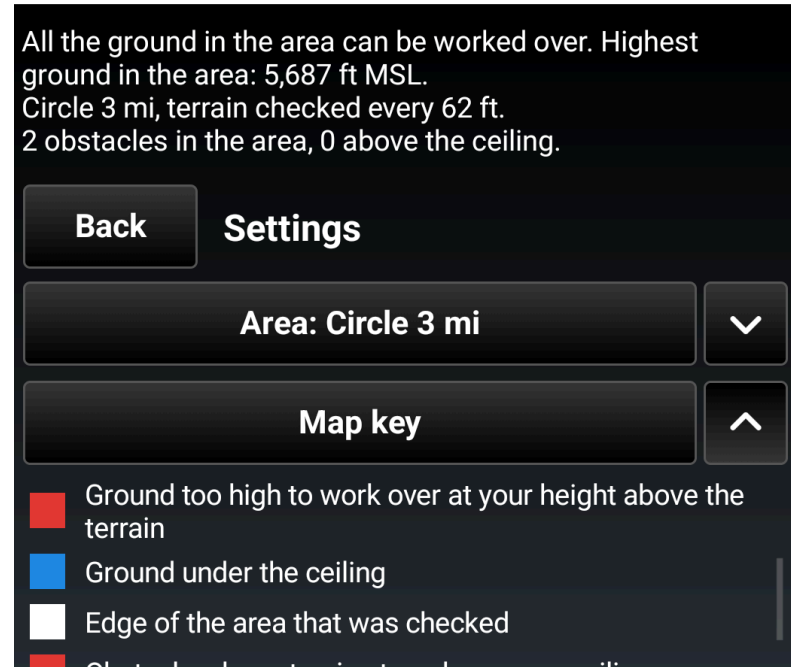
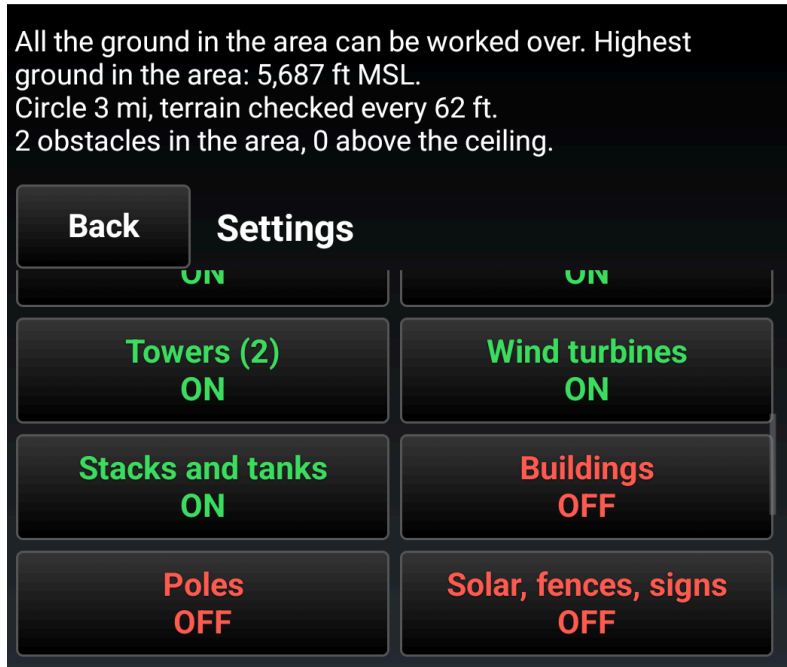


Height above terrain: 400 ft



Back returns to the plan. Here are the obstacle switch, **Taller than**, **Types**, the **Height above terrain** you fly at, the separation, and the size of the circle.

Types and the map key



Types picks which kinds of obstacle are drawn and listed. **Map key** names every color the planner puts on the map.

When there is no terrain data

Only SRTM3 covers this ground. That is too coarse to find a ridge between its posts, so nothing is painted. Load DTED2 covering this ground (the Map Depot plugin can download it), then place the launch point again.

Islands OFF

Settings

Launch point

Ground here: unknown until DTED2 covers it.

CEILING

Given a ceiling

Need a ceiling

OBSTACLES IN THE AREA, NEAREST TO YOU FIRST (43)

The planner reads the elevation ATAK carries. Where it is missing, the plugin says so and leaves that ground unpainted rather than computing a ceiling from data it does not have.

Load the elevation for the area and the readouts fill in.

When there is no network

Islands cover 14% of the area. Highest ground in the area:
5,687 ft MSL.
Circle 3 mi, terrain checked every 62 ft.
2 obstacles in the area, 2 above the ceiling. No answer from
the FAA; showing what this phone saved for this area.

Islands ON

Settings

Launch point

The status line says what you are looking at in words, and names what it could not reach. A failed fetch never empties the map.

Filing an authorization

File an authorization

LAANC in Aloft Air Control

Another FAA-approved supplier

FAA DroneZone (web) - what LAANC does not cover

File LAANC asks which way you are filing and takes you there.

LAANC is granted through FAA-approved suppliers, not by the FAA directly, so the button opens the supplier's app if you have it and its website if you do not. Anything LAANC does not cover goes through FAA DroneZone instead, which is the second choice on the dialog.

AirAware does not file anything for you and does not send it your plan. It takes you to the form.

Notifying Flight Service

Notify Flight Service

S operation - notice to airmen

Position: 33.84888, -117.61392

Altitude: 5,280 ft radius

Altitude: surface to 200 ft AGL (2000 ft MSL)

Duration: from now until you say otherwise

Purpose: UAS fire mapping

Go to 1800wxbrief.com: UAS tab, Planning, UAS
TAM Form.

or phone to Flight Service: 1-877-487-6867

Copy

Call

File online

Notify Flight Service gathers the plan's own numbers, ready to read out or paste: where you are launching, the area, the ceiling, and when.

Nothing is transmitted. You place the call or open the form.

This manual

This guide is inside the plugin. Open it from **Settings**, then **Tool Preferences**, then **AirAware**, then **Plugin Documentation**.

Worth knowing

- **Advisory only.** Everything AirAware draws is a picture of published data at the time it reached this phone. It is not a clearance, it is not current by itself, and it does not replace charts, NOTAMs or a call.
- **Nothing is shared.** No TAK Server, no Data Sync, no CoT. What you switch on is on your device only, and your position is never sent anywhere.
- **What it holds, it keeps.** Airspace, restrictions, obstacles and UAS ceilings that have reached this phone come back after a restart with no network. METARs do not, on purpose.
- **It says what it is not showing.** A trimmed list, a layer under its zoom gate or an area that was never downloaded is stated in words on the status line. A quiet map is not the same as an empty sky.
- **Data** is published by the Federal Aviation Administration. Use of FAA data does not imply FAA endorsement.